Chapter 13 Warm-Up: A 2D Vector Class

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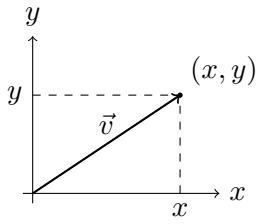
October 26, 2026

1 What Is a Vector?

A 2D vector has an x-component and a y-component:

$$\vec{v} = \langle x, y \rangle.$$

Geometrically, we can think of it as an arrow from the origin to the point (x, y).



2 Magnitude and Angle

The **magnitude** of a vector is its length:

$$|\vec{v}| = \sqrt{x^2 + y^2}.$$

The **angle** θ it makes with the positive x-axis is

$$\theta = \tan^{-1}\left(\frac{y}{x}\right).$$

3 Build the Class and Add Methods

We can represent a vector using a class:

```
class Vector2D:
    def __init__(self, x, y):
        self.x = x
        self.y = y
```

Begin your file with

```
import math
```

Then add a magnitude method:

```
def magnitude(self):
    return math.sqrt(self.x**2+self.y**2)
```

Add an angle method:

```
def angle(self):
    return math.atan2(self.y,self.x)
```

What should

```
v1.magnitude()
```

return for $\langle 3, 4 \rangle$? _____

4 Add and Subtract Vectors

For

$$\vec{u} = \langle x_1, y_1 \rangle, \qquad \vec{v} = \langle x_2, y_2 \rangle,$$

vector addition is

$$\vec{u} + \vec{v} = \langle x_1 + x_2,\ y_1 + y_2 \rangle$$

and subtraction is

$$\vec{u} - \vec{v} = \langle x_1 - x_2,\ y_1 - y_2 \rangle.$$

Add these methods:

```
def __add__(self, other):
    return Vector2D(
        self.x+other.x,self.y+other.y)
def __sub__(self, other):
    return Vector2D(
        self.x-other.x,self.y-other.y)
```

Test your addition and subtraction methods with

```
v1 = Vector2D(3, 4)
v2 = Vector2D(1, -2)
```

5 Looking Ahead: Inheritance

A complex number $z = a + jb$ also has two components, much like a 2D vector. How might you create a class `ComplexNumber` that **inherits** from `Vector2D`? What parts could it reuse? What new methods or meanings would it need? **Hint:** A child class can begin like this:

```
class ComplexNumber(Vector2D):
    ...
```